MIL-M-0038510/3E(USAF)
1 December 1975
USED IN LIEU OF
MIL-M-38510/3B
23 December 1971
SUPERSEDING
MIL-M-0038510/3D(USAF)
1 May 1974

MILITARY SPECIFICATION

MICROCIRCUIT, DIGITAL, TTL, NAND
BUFFER, MONOLITHIC SILICON

This limited coordination military specification has been prepared by the Air Force Space and Missile Systems Organization based upon currently available technical information, but it has not been approved for promulgation as a coordinated revision of Military Specification MIL-M-38510/3B. It is subject to modification. However, pending its promulgation as a coordinated military specification, it may be used in procurement.

1. SCOPE

1.1 Scope. This specification covers the requirements for a monolithic silicon, TTL, positive NAND buffer microcircuit for use in Air Force missile and space systems.

1.2 Part number. The complete part number shall be as shown in the following example:

M38510	/003	01	S	T	D
Military designator	Detail specification	Device type (1.2.1)	Device class (1.2.2)	Case outline (1.2.3)	Lead finish (3.3)

1.2.1 Device type. The device type shall be as shown in the following:

Device type	Circuit	Similar commercial type
01	Dual, 4-input positive NAND buffer	5440
02	Quad, 2-input positive NAND buffer	5437
03	Quad, 2-input positive NAND buffer (open collector outputs)	5438

1.2.2 Device class. The device class shall be class S as defined in MIL-M-0038510.

1.2.3 Case outline. The case outline shall be as specified in figure 10 of MIL-M-0038510 and shall be identified as outline T.

1.3 Absolute maximum ratings.

Supply voltage range	-0.5 Vdc to 7.0 Vdc
Input voltage range	0 to 5.5 Vdc
Storage temperature range	-65°C to 150°C
Lead temperature (soldering, 10 seconds)	300°C

1.4 Recommended operating conditions.

Ambient operating temperature range	-55°C to 125°C
Supply voltage	4.5 Vdc minimum to 5.5 Vdc maximum
Minimum high level input voltage	2.0 Vdc
Maximum low level input voltage	0.8 Vdc

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2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following document, of the issue in effect on date of invitation for bids or request for proposal, forms a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-0038510 - Microcircuits, General Specification for.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-0038510 and as specified herein.

3.2 Design, construction, weight and physical dimensions. The design, construction, weight and physical dimensions shall be as specified in MIL-M-0038510 and herein.

3.2.1 Logic diagram and terminal connections. The logic diagram and terminal connections shall be as specified on figure 1.

3.2.2 Truth tables and logic equations. The truth tables and logic equations shall be as specified on figure 2.

3.2.3 Schematic circuit. The schematic circuit shall be as specified on figure 3.

3.2.4 Case outline. The case outline shall be as specified in 1.2.3.

3.2.5 Weight. The maximum weight shall be 1 gram.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-0038510.

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in tables I and II.

3.4.1 Delta drift limits. The delta drift limits shall be as specified in table I.

3.5 Screening. Screening shall be in accordance with table I of MIL-M-0038510 and as specified herein.

3.5.1 Interim electrical parameters. The interim electrical parameters shall consist of all the electrical characteristics tests as specified in table I. Each test shall be performed go, no-go at -55°C and 125°C. Each test shall also be read and recorded at 25°C. For all tests, the devices shall meet the limits specified in table I.

3.5.2 Final electrical tests. The final electrical tests shall consist of all the electrical characteristics tests specified in table I. Each test shall be read and recorded at -55°C, 25°C and 125°C. For all tests, the devices shall meet the limits specified in table I.

3.5.2.1 Delta computation. Delta computations shall be performed for those parameters in table I which are marked with an asterisk. All devices shall meet the applicable delta limits specified in table I. (See section 6 in MIL-M-0038510.)

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TABLE I. Electrical characteristics tests (device type 01).

Test No. 1/ 2/	Characteristics	Symbol	Conditions	Limits 3/		Delta (Δ) 1/	Units
				Min	Max		
1	Minimum high level input voltage	V_{IH}	V_{CC} = 4.5 V	2		-	V
2	Maximum low level input voltage	V_{IL}	V_{CC} = 4.5 V		0.8	-	V
3*	High level output voltage	V_{OH}	V_{CC} = 4.5 V, V_{IL} = 0.8 I_{OH} = -1.2 mA	2.4		±.12	V
4*	Low level output voltage	V_{OL}	V_{CC} = 4.5 V, V_{IH} = 2 V I_{OL} = 48 mA		0.4	±.04	V
5*	High level input current (each input)	I_{IH1}	V_{CC} = 5.5 V, V_{IN} = 2.4 V		40	10X or 4	µA
		I_{IH2}	V_{CC} = 5.5 V, V_{IN} = 5.5 V		100	N/A	µA
6	Low level input current (each input)	I_{IL}	V_{CC} = 5.5 V, V_{IN} = 0.4 V	-0.7	-1.6	-	mA
7	Short circuit output current	I_{OS}	V_{CC} = 5.5 V, V_{IN} = 4.5 V V_{IL} = 0	-20	-70	-	mA
8	Supply current	I_{CCL}	V_{CC} = 5.5 V, V_{IN} = 5.5 V		27	-	mA
9	Supply current	I_{CCH}	V_{CC} = 5.5 V, V_{IL} = 0 V		8	-	mA
10	Input clamp voltage	V_{IC}	V_{CC} = 4.5 V, I_{IC} = -12 mA T_A = 25°C		-1.5	-	V

1/ Delta computation shall be performed on parameters marked with an asterisk (*) with data read and recorded ("10X" denotes "10 times the initial measurements"). Where two delta values are indicated for one parameter, the greater value shall apply.
2/ Test sequence is optional with the manufacturer.
3/ Unless otherwise specified, the limits apply at T_A = -55°C to +125°C.

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TABLE I. Electrical characteristics tests (device type 02).

Test No. 1/ -/	Characteristics	Symbol	Conditions	Limits 3/		Delta (Δ) 1/	Units
				Min	Max		
1	Minimum high level input voltage	V_{IH}	V_{CC} = 4.5 V	2		-	V
2	Maximum low level input voltage	V_{IL}	V_{CC} = 4.5 V		0.8	-	V
3*	High level output voltage	V_{OH}	V_{CC} = 4.5 V, V_{IL} = 0.8 I_{OH} = -1.2 mA	2.4		±.12	V
4*	Low level output voltage	V_{OL}	V_{CC} = 4.5 V, V_{IH} = 2 V I_{OL} = 48 mA		0.4	±.04	V
5*	High level input current (each input)	I_{IH1}	V_{CC} = 5.5 V, V_{IN} = 2.4 V		40	4.0	µA
		I_{IH2}	V_{CC} = 5.5 V, V_{IN} = 5.5 V		100	N/A	µA
6	Low level input current (each input)	I_{IL}	V_{CC} = 5.5 V, V_{IN} = 0.4 V	-0.4	-1.6	-	mA
7	Short circuit output current	I_{OS}	V_{CC} = 5.5 V, V_{IL} = 0	-20	-70	-	mA
8	Supply current	I_{CCL}	V_{CC} = 5.5 V, V_{IN} = 5.5 V		54	-	mA
9	Supply current	I_{CCH}	V_{CC} = 5.5 V, V_{IL} = 0 V		15.5	-	mA
10	Input clamp voltage	V_{IC}	V_{CC} = 4.5 V, I_{IC} = -12 mA T_A = 25°C		-1.5	-	V

1/ Delta computation shall be performed on parameters marked with an asterisk (*) with data read and recorded.
2/ Test sequence is optional with the manufacturer.
3/ Unless otherwise specified, the limits apply at T_A = -55°C to +125°C.

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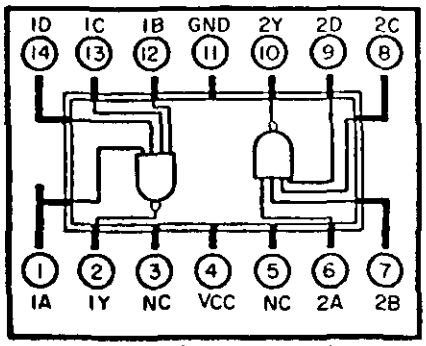
TABLE I. Electrical characteristics tests (device type 03).

Test No. 1/ 2/	Characteristics	Symbol	Conditions	Limits 3/		Delta (Δ) 1/	Units
				Min	Max		
1	Minimum high level input voltage	V_{IH}	V_{CC} = 4.5 V	2		-	V
2	Maximum low level input voltage	V_{IL}	V_{CC} = 4.5 V		0.8	-	V
3	Maximum collector cut-off current	I_{CEX}	V_{CC} = 4.5 V, V_{OH} = 5.5 V V_{IL} = 0.8		250	-	µA
4*	Low level output voltage	V_{OL}	V_{CC} = 4.5 V, V_{IH} = 2 V I_{OL} = 48 mA		0.4	±.04	V
5*	High level input current (each input)	I_{IH1}	V_{CC} = 5.5 V, V_{IN} = 2.4 V		40	±4.0	µA
		I_{IH2}	V_{CC} = 5.5 V, V_{IN} = 5.5 V		100	N/A	µA
6	Low level input current (each input)	I_{IL}	V_{CC} = 5.5 V, V_{IN} = 0.4 V	-0.7	-1.6	-	mA
7	Supply current	I_{CCL}	V_{CC} = 5.5 V, V_{IN} = 5.5 V		54	-	mA
8	Supply current	I_{CCH}	V_{CC} = 5.5 V, V_{IL} = 0 V		8.5	-	mA
9	Input clamp voltage	V_{IC}	V_{CC} = 4.5 V, I_{IC} = -12 mA T_A = 25°C		-1.5	-	V

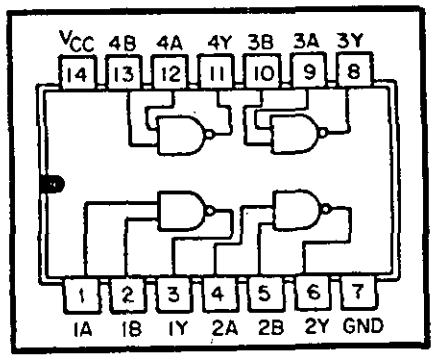
1/ Delta computation shall be performed on parameters marked with an asterisk (*) with data read and recorded.
2/ Test sequence is optional with the manufacturer.
3/ Unless otherwise specified, the limits apply at T_A = -55°C to +125°C.

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Device Type 01



Device type 02 and 03



NC-No Internal Connection

FIGURE 1. Logic diagram and terminal connections.

Device type 01

Truth table				
Input				Output
A	B	C	D	Y
L	L	L	L	H
H	L	L	L	H
L	H	L	L	H
H	H	L	L	H
L	L	H	L	H
H	L	H	L	H
L	H	H	L	H
H	H	H	L	H
L	L	L	H	H
H	L	L	H	H
L	H	L	H	H
H	H	L	H	H
L	L	H	H	H
H	L	H	H	H
L	H	H	H	H
H	H	H	H	L

Positive logic $Y = \overline{ABCD}$

Device types 02 and 03

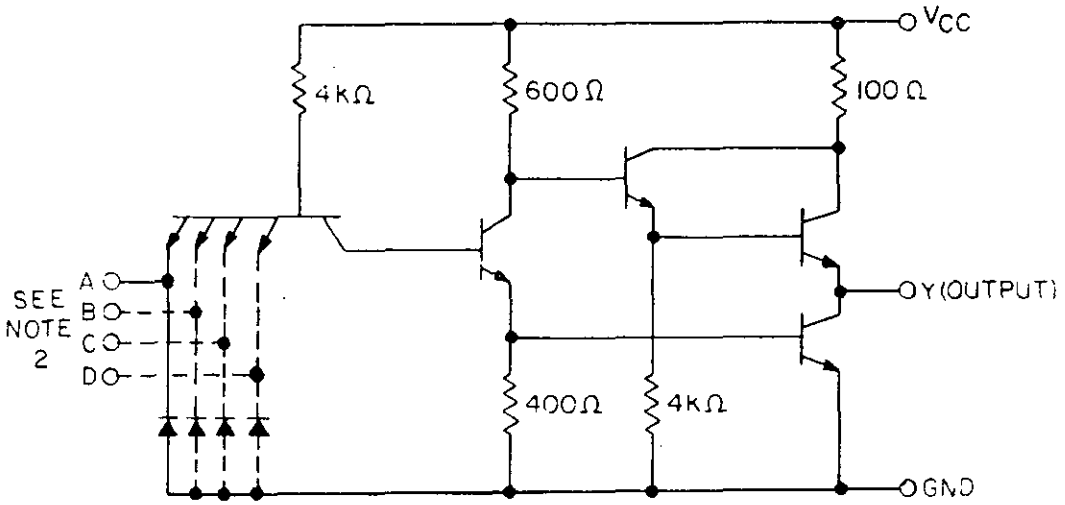
Truth table each gate		
INPUT		OUTPUT
A	B	Y
L	L	H
H	L	H
L	H	H
H	H	L

Positive Logic $Y = \overline{AB}$

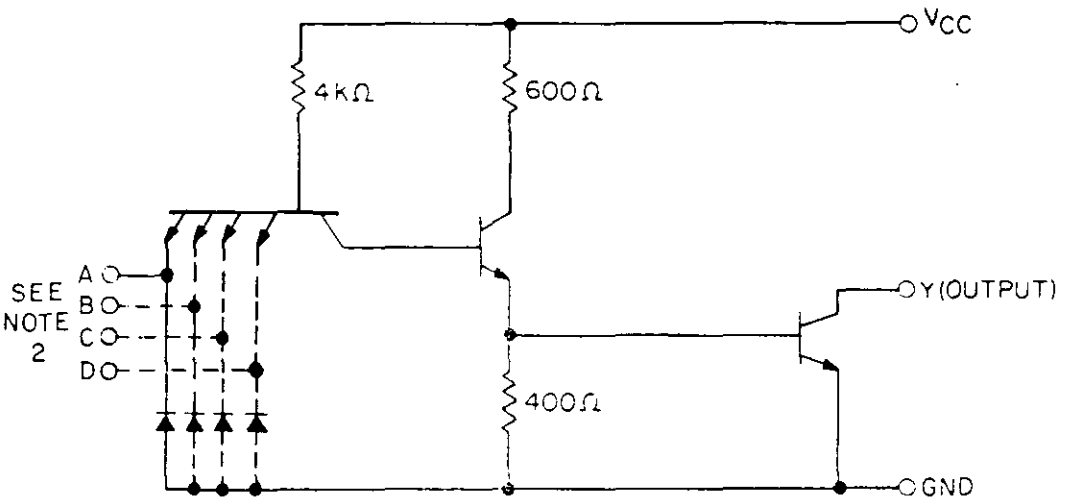
FIGURE 2. Truth tables.

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Device types 01 and 02



Device type 03

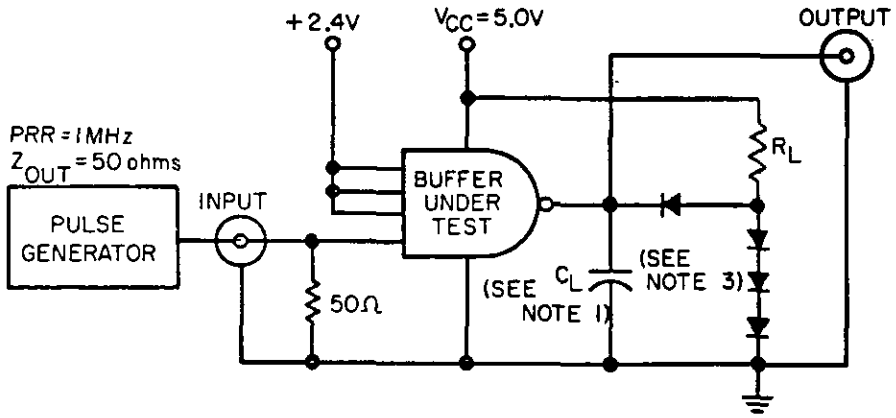


NOTES:
1. All resistance values shown are nominal.
2. Number of inputs is determined by the logic configuration shown in figure 1.

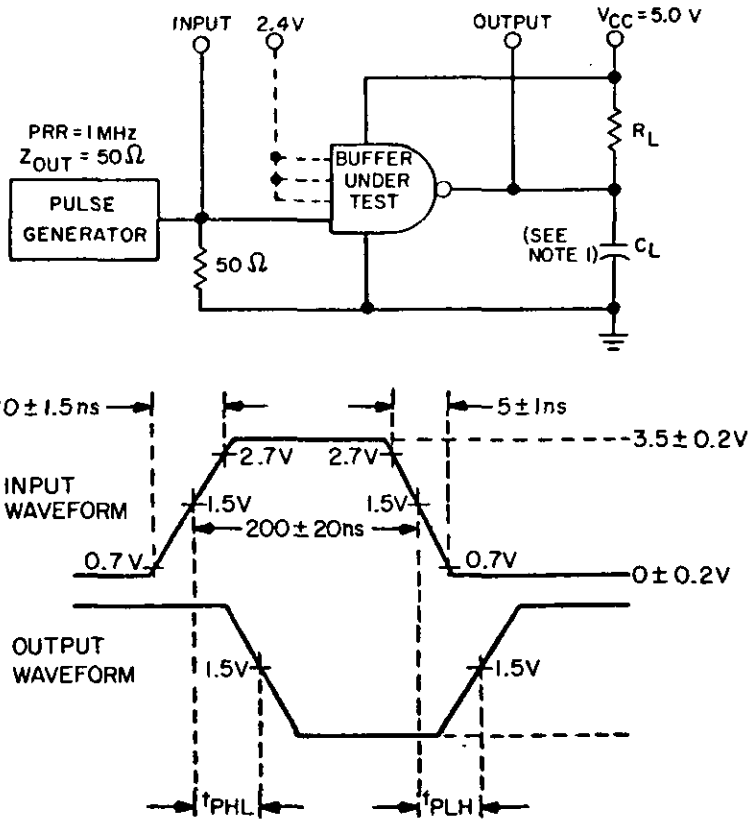
FIGURE 3. Schematic circuits

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Device types 01 and 02



Device type 03



NOTES:
1. Including scope probe, wiring, and stray capacitance, without package in test fixture, C_L = 125 pF ±10%.
2. Voltage measurements are to be made with respect to network ground terminal.
3. All diodes are 1N3064 or equivalent.
4. R_L = 133Ω ± 5%.

FIGURE 4. Switching time test circuit.

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TABLE II. Switching characteristics (device types 01, 02, 03).

Characteristics	Symbol	Conditions	Test figure	Temp.	Limits		Units
					Min	Max	
Propagation delay time, high-to-low level	t_{PHL}	V_{CC} = 5.0 V C_L = 125 pF R_L = 133Ω	4	25°C 125°C -55°C	3 3 3	25 30 30	ns
Propagation delay time, low-to-high level	t_{PLH}			25°C 125°C -55°C	3 3 3	22 27 27	ns

3.5.3 Switching characteristics tests. The switching characteristics tests specified in table II and figure 4 shall be performed at 25°C. All devices shall meet the limits specified in table II.

3.6 Extended life tests. Extended life tests shall be performed in accordance with MIL-M-0038510.

3.7 Marking. Marking shall be in accordance with MIL-M-0038510 and 1.2 herein.

4. PRODUCT ASSURANCE PROVISIONS

4.1 Product assurance provisions. Product assurance shall be in accordance with MIL-M-0038510.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-0038510.

6. NOTES

6.1 Notes. Notes are as specified in MIL-M-0038510.

Custodian:
Air Force - 19

Review activities:
Air Force - 17, 19
DSA - ES

Preparing activity:
Air Force - 19

Agent:
DSA - ES

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